

### Problem 6.6

(a) For the following cases, calculate the value of  $r$  at which the radial probability density of the hydrogen atom reaches its maximum: (i)  $n = 1, l = 0, m = 0$ ; (ii)  $n = 2, l = 1, m = 0$ ; (iii)  $l = n - 1, m = 0$ .

(b) Compare the values obtained with the Bohr radius for circular orbits.

• Now, the problem asks about the radial probability so that means we are looking at  $R_{nl}(r)$  only:

$a_0$ : Bohr radius

$$\psi_{nlm}(r, \theta, \phi) = \underline{R_{nl}(r)} Y_l^m(\theta, \phi)$$

(c) Let's look at how to interpret what is being asked. We start with  $\langle n, l, m | n, l, m \rangle = 1$ .

So,

$$\langle n, l, m | n, l, m \rangle = \underbrace{\left( \int r^2 |R_{nl}|^2 dr \right)}_{\text{Radial}} \underbrace{\left( \int_0^{2\pi} \int_0^\pi |Y_l^m(\theta, \phi)|^2 \sin(\theta) d\theta d\phi \right)}_{\text{Angular}}$$

Now, the probability density is given by  $\mathcal{P}(r)$  in

$$\int \mathcal{P}(r) dr = \int r^2 |R_{nl}(r)|^2 dr$$

Thus, the Radial probability density is

$$\ast \boxed{\mathcal{P}_{nl}(r) = r^2 |R_{nl}(r)|^2} \ast$$

We can use this to solve the problem

(i)  $n = 1, l = 0, m = 0$

From the table we read off  $R_{10}$ :

$$\mathcal{P}_{10}(r) = r^2 |R_{10}(r)|^2 = r^2 \left| 2a_0^{-3/2} e^{-r/a_0} \right|^2$$

$$P_{10}(r) = 4r^2 a_0^{-3} e^{-2r/a_0}$$

- Recall from calc that we can find the max by looking at the functions critical points:

$$\left. \frac{dP_{10}(r)}{dr} \right|_{r=r_1} = 0$$

$$\begin{aligned} \frac{dP_{10}(r)}{dr} &= (8r_1) a_0^{-3} e^{-2r_1/a_0} + 4r_1^2 a_0^{-3} \left( \frac{-2}{a_0} \right) e^{-2r_1/a_0} \\ &= e^{-2r_1/a_0} 8a_0^{-3} (r_1 - r_1^2/a_0) \\ &= 0 \end{aligned}$$

and so we find

$$r_1 (1 - r_1/a_0) = 0$$

or

$$\begin{cases} r_1 = 0 \\ r_1 = a_0 \end{cases}$$

The first is a trivial solution since this would mean  $\psi_{100} = 0$  so we ignore this possibility. The second is the correct answer

$$r_1 = a_0$$

(ii)  $n=2, l=1, m=0$ :

Here we do the exact same thing:  
 $\psi_{210} = R_{21} Y_1^0$

$$P_{21}(r) = r^2 \underbrace{|R_{21}(r)|^2}$$

read off table

$$P_{2,1}(r) = r^2 \left| \left( 3^{-1/2} (2a_0)^{-3/2} \frac{r}{a_0} e^{-r/2a_0} \right) \right|^2$$

$$P_{2,1}(r) = \frac{1}{24a_0^5} r^4 e^{-r/a_0}$$

To find the max:

$$\frac{dP_{2,1}}{dr} \Big|_{r=r_2} = 0 \Rightarrow 4r_2^3 - \frac{r_2^4}{a_0} = 0$$

$$r_2 = 4a_0$$

litc)  $n, l = n-1, m=0$

$$\psi_{n(n-1)0} = R_{n(n-1)}(r) Y_{n-1}^0(\theta, \phi)$$

$$P_{n(n-1)}(r) = r^2 |R_{n(n-1)}(r)|^2$$

→ Use Eq. (7.4.1) of notes

→ Answer should be

$$r_n = n^2 a_0$$

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⑥ (Solve this).